

Projection Mapping Method Using Projector-LiDAR (light detection and ranging) Calibration

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ABSTRACT

The projector-LiDAR system performs projection mapping on a long-distance object after calibrating the projector using LiDAR. With LiDAR equipped with an RGB camera, users can calibrate the projector in just two scans. Because this system knows the ray formula for every pixel shooting from the projector, it can make accurate projections. In particular, the system introduces how to perform linear interpolation in the process of obtaining a formula and a methodology that allows users to immediately know the 3D coordinates of a desired part in a 2D image. Through a long-distance experiment, the projection mapping result of the proposed method showed an error of less than 5mm, and the existing system had a maximum error of 30mm.

Keywords: projection mapping, virtual worlds, calibration, LiDAR

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Mixed / augmented reality

1 INTRODUCTION

Projection mapping is widely used in the field of augmented reality (AR) and mixed reality (MR) as it can blend virtual and real elements together [2]. Space augmentation through projection has the advantage that many people can experience the same experience at the same time without wearing an HMD, so it is effective for large-scale, long-distance projection [4]. Projection mapping projects require more advanced technology as the size and complexity of the objects increase [3, 5]. The projector-camera system (procam) is a method that was mainly used for projection mapping. This system is mainly effective when applied at short distances, and projection is not accurate when the mapping target is long distance or non-planar. This is because the calibration results vary depending on factors such as perspective distortion, lens distortion, and images used for projector calibration, which can result in errors in the projection mapping. Additionally, traditional calibration methods require pattern images to be taken four or more times from various angles, making the process tedious and requiring re-calibration every time the projector position is changed. The 'projector-LiDAR system' we propose uses LiDAR-based calibration to accurately and robustly project objects at long distances, large scales, and non-planar surfaces. The calibration we refer to is a projector system that directly calculates the direction of the projector's light using LiDAR. This system is practical in situations where long-distance projection is required because it is free from the distortion of the projector lens because

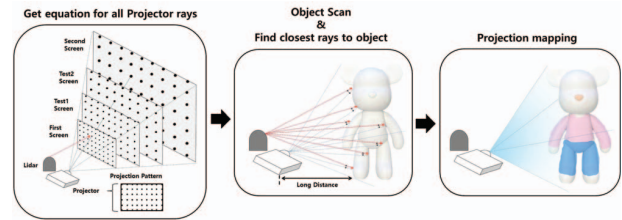


Figure 1: Overview of the calibration and projection process.

it calculates the light equation of every pixel of the projector. For calibration, it is also convenient because only two scans are required from an angle other than multiple angles.

2 METHOD

The projector-LiDAR system is based on the concept of knowing the projector's ray equation (PRE) for all the projector's rays by reading the 3D position of the light from the projector on the screen with LiDAR. PRE for all the projector's rays is determined by performing a total of two LiDAR scans. At this time, a 1920x1080 image with 9 columns and 7 rows of dots is projected onto the screen (Fig. 1). The spacing between dots in each column is 240px, and the spacing in rows is 180px. The first LiDAR scan placed the screen close to the projector, and the second placed the screen further away from the projector. After scanning, the resulting point cloud is transformed into a 2D image using a spherical coordinate system. This allows for accurate projection as the LiDAR scanner measures while rotating and the camera inside the scanner captures equirectangular images, making projection in a spherical coordinate system intuitive. Therefore, a 2D image is obtained by using the ϕ and θ values obtained from the spherical coordinate system, allowing to know the 3D coordinates of a point when selected in the 2D image. A total of two 2D images were obtained by converting two LiDAR scans. To find the PRE, Harris corner detection is used on each image to locate the center values of the pattern [1]. By finding the 3D coordinates of these center values, the equation of a straight line in space can be created. Therefore, a total of 63 equations of a 3D straight line that can be obtained directly are obtained (Fig. 2). Then interpolation between the 63 line segments can calculate the 2,073,600 equations for every pixel in the projector. To check if the equations of the straight line found are correct, set up screens like the test1 screen and test2 screen in Fig. 1. And after obtaining the 3D coordinates of the test, users can see if it passes the equation of the straight line obtained previously.

After calculating the PRE, the user scans the Desired projection location or where the target object is located with LiDAR. Once 3D point for the target is found, we compare it with the PRE to see which number of PRE matches the point. If there is no exact match, the PRE closest to the 3D point is used.

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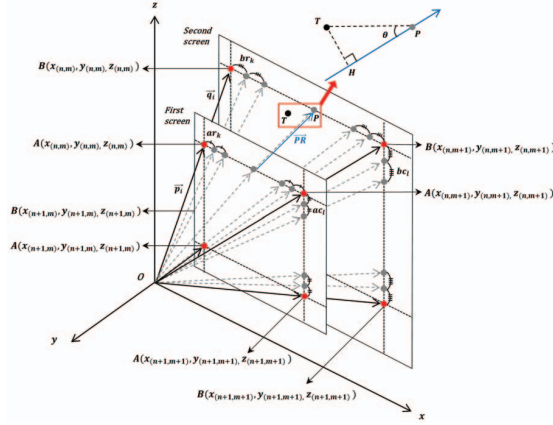


Figure 2: Variables used when creating PRE and linear interpolation with first screen and second screen tests. $A(x(n,m), y(n,m), z(n,m))$ is the (x, y, z) coordinates of n rows and m columns on the first screen. $B(x(n,m), y(n,m), z(n,m))$ is the (x, y, z) coordinates of n rows and m columns on the second screen. $\vec{p}_i$ is the ray vector of all pixels on first screen. $\vec{q}_i$ is the ray vector of all pixels on second screen. Every pixel spacing in a row of screen A is ar_k . Every pixel spacing in a column of screen A is ac_l . Every pixel spacing in a row of screen B is br_k . Every pixel spacing in a column of screen B is bc_l . The straight line connecting the coordinates in the same row and column on screen A and screen B is PR_l . The coordinates of the target to be mapped are T , the distance from T to the nearest pixel ray is $|TH|$.

3 RESULTS AND DISCUSSION

The hardware used for the test was a Leica BLK360 LiDAR scanner (13 Mpixel 4-camera system, $360^\circ \times 270^\circ$ spherical image), an Optoma EH502 projector (resolution: 1920×1080 pixels, display frequency: 144Hz) and a Canon EOS 650D (resolution: 5184×3456 pixels). The projector, LiDAR and camera were placed side by side horizontally. The experimental bear model size is $35\text{cm} \times 25\text{cm} \times 70\text{cm}$. The zoom and direction conditions of the projector in the projector-LiDAR test and the procam test are all the same. Projection mapping experiments were conducted in three locations. In the small room, the first place, the projector and the test object were 4.43m apart, the distance between the projector and the first screen was 1.8m, and the distance between the second screen was 4.43m. The second is a large room, the distance between the projector and the test subject is 10m, the distance from the first screen is 2.7m, and the distance from the second screen is 4.32m. In the last corridor, the distance between the projector and the test subject is 30m. The first screen in the projector was 1.8m away, and the second screen was 7.7m away. The procam and projector-LiDAR were tested at every location and the projection mapping results at each location were compared. Projection error measured the pixel error and physical error of the mapping, i.e., it is to see if the red line is accurately projected on the edge of the bear model. The error measurement measured how far the location to be mapped was off in pixels and measured how many centimeters it was physically off.

In the case of the existing procam method, the pixel error in the small room was 3px based on the arm of the bear model, and the physical error was about 10mm (Fig. 3a). When conducted in a large room, the projection error occurred by 6px, and the actual physical error was 30mm (Fig. 3b). In an experiment in the corridor, which is the longest street, the camera-based projection did not even come out in the form of a bear model, so it was not possible to project (Fig. 3c). On the other hand, the projection mapping of the projector-LiDAR calibration showed high mapping accuracy in all experimental sites.

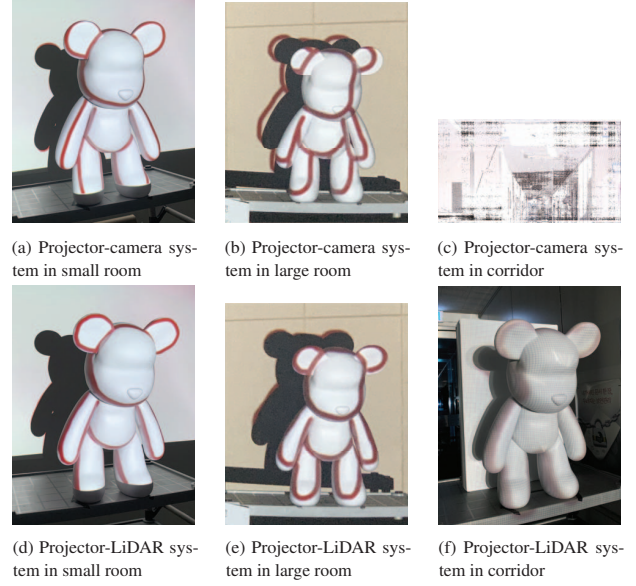


Figure 3: Results of the projection mapping.

In the small room and large room, the actual physical error was less than 1mm and the pixel error was less than 1px (Fig. 3d,3e). In the hallway experiment, the pixel error at the model border was less than 2px and the physical error was less than 5mm (Fig. 3f). This proves that our proposed 'projector-LiDAR system' performs robust projection mapping regardless of distance, size, and volume. Even if the target for projection mapping is largescale and long-distance, there is an advantage in that mapping is possible with little errors by extracting only the 3D coordinates of the target through LiDAR scanning.

As a future work, if a real time LiDAR is used, projection mapping can be performed immediately whenever the location of the projection target changes or the object itself changes. In addition, the fact that 3D coordinate values of a location selected in a 2D image can be received can be utilized in a feature point matching algorithm. The two point clouds to be matched are transformed into images, and image matching is performed to find matching points. Point cloud matching using only these points will save time.

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